Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Full-stack software developer with 2+ years building **JavaScript/TypeScript interfaces**, Python/Java services, SQL Server data systems, and internal business applications; experienced in OOP, MVC, REST APIs, refactoring, testing, documentation, and AI-assisted development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Developed **server-side Python services and relational data structures** for 500+ users across SQL Server and Azure SQL, refactoring indexed queries and access patterns to reduce latency from 1.2 seconds to 350 milliseconds.
- Built and supported **robust production integration workflows** across 5,000+ monthly jobs, adding validation, reconciliation, error handling, technical documentation, and CI/CD releases to achieve 99%+ reliability for internal applications.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Created **client-side React and JavaScript interfaces** and Flask server code for uploads, processing, history, and reporting, delivering responsive prototypes and production features supporting 150+ users and 10,000+ daily requests.
- Integrated **LLM, vision, and OCR capabilities** through adaptable REST APIs, adding validation, usage logging, fallback handling, and concurrent execution to prototype engineering workflows quickly while reducing turnaround time 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Refactored **object-oriented API and data components** using Singleton and Factory patterns, SQL, caching, AWS Glue, and S3, improving response times 30%, pipeline efficiency 68%, storage costs 15%, and maintainability.
- Collaborated on **requirements, code reviews, integration, and testing** with Microsoft and GitHub stakeholders, automating builds, deployments, and 20+ tests through Jenkins, PyTest, and Mockito across Windows, Linux, and macOS.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **React, JavaScript, Node.js, and Azure SQL business applications** across eight services for authentication, records, billing, permissions, and notifications, synchronizing 2M+ records through REST APIs and real-time WebSockets.
- Maintained **database-backed production applications** through Jest, Selenium, Sentry, Azure Monitor, Docker, and Jenkins CI/CD, debugging defects and cutting release time 50% while improving performance, reliability, and process consistency.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **React and TypeScript client interfaces** with responsive HTML/CSS, Spring Boot and Flask server APIs, and Firebase synchronization, delivering booking, pricing, and status workflows across 50,000+ records.
- Translated client requirements into **cross-platform application features**, mapping 12+ integrations and delivering unit-tested, version-controlled enhancements that improved compatibility 15%, reduced design errors 25%, and accelerated project delivery 22%.

Skills

Languages: JavaScript, TypeScript, Python, Java, C++, SQL, Bash, HTML, CSS
Full Stack: React.js, Next.js, Node.js, Express.js, FastAPI, Flask, Spring Boot, MVC, REST APIs, WebSockets
Data: SQL Server, Azure SQL, PostgreSQL, MySQL, MongoDB, Firebase, relational design, indexing, data visualization
AI: LLMs, RAG, vector search, prompt engineering, PyTorch, Transformers, FAISS, AI-assisted development
Engineering: OOP, refactoring, testing, debugging, Git, Jenkins, GitHub Actions, CI/CD, Azure, Docker, documentation

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Enhanced a **cross-platform database application** across Java and TypeScript modules, supporting 10+ SQL/NoSQL systems with schema exploration, query generation, data management, Docker, Maven, and reusable desktop development workflows.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Built a **RAG and vector-search application** processing 50,000+ documents with incremental indexing, metadata persistence, secure access, collaborative workflows, and LLM summarization, reducing retrieval latency 40% and review time 60%.

HMIS Live | *TypeScript, REST APIs, Sentry, TailwindCSS*
- Developed a **full-stack internal operations application** with TypeScript dashboards, REST integrations, authentication, role-based workflows, notifications, and Sentry monitoring, halving record retrieval time while reducing administrative workload 30%.